Achieving Robust Out-of-Distribution Generalization in Peripheral Blood Smears via Custom Attention Mechanisms, Medical Enhanced Filtering, and Inference-Time Domain Adaptation

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May 17, 2026

Abstract

Deep learning models for peripheral white blood cell (WBC) classification frequently exhibit severe performance degradation when exposed to out-of-distribution (OOD) data arising from differences in microscopy hardware and staining protocols, limiting their clinical applicability. We propose an end-to-end WBC classification system designed specifically for robust generalization and high computational efficiency. Built upon a DenseNet121 backbone [1], the architecture integrates three original contributions: (1) a deterministic five-step *Medical Enhanced Filter* (MEF) preprocessing pipeline that standardizes cross-device image variability before feature extraction; (2) a CBAM-style *WBCAttention* mechanism [2] employing sequential channel and spatial refinement to suppress background artifacts; and (3) a *MedSwish* activation function with trainable parameters that preserves gradient flow in negative regions [3]. The system achieves **98.53%** in-distribution accuracy on TestA while maintaining a parameter footprint of only **7.83 M**, yielding inference latency of **14.2 ms**. Crucially, the system demonstrates strong resilience to domain shift, reaching **89.05%** OOD accuracy on the TestB dataset—a **+32.09 percentage point** improvement over the unadapted baseline—through a retraining-free inference-time adaptation pipeline combining binary routing, Reinhard stain normalization [4], and lightweight test-time augmentation [5]. To prevent shortcut learning [6], a dynamic Explainability Guardrail integrates Grad-CAM [7] focus ratio monitoring directly into the training loop. An autonomous multi-modal LLM agent further interprets heatmaps post-hoc to generate clinical insights. Ablation studies validate each architectural component and confirm that the proposed MEF variant outperforms both plain and morphologically augmented preprocessing alternatives on OOD data.

Keywords: white blood cell classification, out-of-distribution generalization, attention mechanism, domain adaptation, explainability, medical image preprocessing, shortcut learning

1 Introduction

Peripheral blood smear analysis is the gold standard for diagnosing a wide spectrum of hematological conditions, including bacterial and viral infections, allergic reactions, anemia, and leukemia. In clinical practice, a trained hematologist manually identifies and counts five leukocyte subtypes—Neutrophil, Eosinophil, Basophil, Lymphocyte, and Monocyte—a process that consumes 15–20 minutes per sample and introduces inter- and intra-observer variability that is well-documented in the literature [8, 9]. High-volume hospitals processing hundreds of samples daily face both throughput bottlenecks and fatigue-related error risk.

Deep learning has demonstrated strong in-distribution performance on this task, yet clinical deployment remains limited by a fundamental challenge: models trained on images from one laboratory routinely fail when presented with images captured by different microscopes, cameras, or staining protocols. This domain shift problem is compounded by the tendency of convolutional networks to exploit spurious background correlations—erythrocyte density, staining artifacts, slide marks—rather than the morphological features of the leukocyte itself [6]. A model that achieves 98% accuracy on a held-out set from the same device may collapse to near-random performance on images from a different device, as our baseline results confirm (56.96% TestB accuracy without adaptation).

This work addresses both challenges simultaneously.

The primary contributions are as follows:

1. **Medical Enhanced Filter (MEF):** A five-step deterministic preprocessing pipeline combining percentile-based exposure normalization, dual-scale adaptive histogram equalization in LAB space, edge-preserving bilateral filtering, morphological nucleus highlighting, and selective weighted sharpening.

2. **WBCAttention:** A CBAM-inspired hybrid attention module adapted for hematological imagery, employing sequential channel and spatial gating to suppress irrelevant background regions.

3. **MedSwish:** A custom activation function with trainable parameters that extends the Swish family with a bounded enhancement term, preserving gradient signal in the negative domain.

4. **Inference-Time Adaptation Pipeline:** A three-stage retraining-free domain adaptation strategy—binary output routing, Reinhard stain normalization, and lightweight test-time augmentation—that raises OOD accuracy from 56.96% to 89.05%.

5. **Dynamic Explainability Guardrail:** An XAI-aware training callback that computes Grad-CAM focus ratios every two epochs and halts training when attention persistently falls outside the cellular foreground.

6. **Production REST API:** A Flask-based web service with input validation, structured logging, and fault-tolerant error handling delivering responses in 200–350 ms, suitable for integration with Hospital Information Systems (HIS).

2 Related Work

2.1 WBC Classification with Deep Learning

Early approaches to automated leukocyte classification relied on hand-crafted features combined with classical classifiers such as SVMs [10]. The transition to deep learning brought transfer-learned CNNs to this domain, achieving high in-distribution accuracy on datasets such as Raabin-WBC and ALL-IDB [11, 12]. However, these studies predominantly report validation accuracy on data drawn from the same acquisition setup as training, leaving generalization to unseen hardware largely unevaluated.

2.2 Domain Shift in Medical Imaging

Stain normalization has been widely studied to address inter-laboratory color variability. Reinhard normalization [4] transfers the LAB-space statistics of a target domain to a source image. Macenko's SVD-based stain separation [13] offers a more principled decomposition but requires careful calibration of reference statistics, as our ablation demonstrates (Section 5.5). Test-time augmentation [5] improves robustness by ensembling predictions across augmented variants of the same input at inference.

2.3 Attention Mechanisms

CBAM [2] introduced sequential channel and spatial attention gates that have become widely adopted in medical image analysis. Our WBCAttention adapts this paradigm specifically to hematological imagery, where the primary challenge is suppressing the abundant background formed by erythrocytes and staining artifacts.

2.4 Shortcut Learning and Explainability

Shortcut learning [6], wherein models exploit spurious correlations predictive only in the training distribution, is particularly acute in medical imaging. Grad-CAM [7] is the standard tool for diagnosing this failure mode post-hoc. Our work integrates Grad-CAM focus ratio measurement directly into the training loop as an early-stopping criterion, converting a diagnostic tool into a preventive mechanism.

3 Dataset

All experiments use the Raabin-WBC dataset [11], an open-source collection released by Tehran University with institutional ethics approval. The dataset comprises Giemsa-stained peripheral blood smear images captured under two distinct acquisition conditions: a professional laboratory microscope camera and a consumer smartphone camera (Samsung Galaxy S5). This hardware heterogeneity is the primary source of the domain shift evaluated in this work.

Classes: Neutrophil, Eosinophil, Basophil, Lymphocyte, Monocyte.

Splits:

- **Train:** ≈12 000 images.
- **TestA:** ≈4 339 images (in-distribution).
- **TestB:** held-out set captured with different hardware (OOD); contains Lymphocyte and Neutrophil classes only.

Class imbalance is pronounced (Neutrophil: 2 660 samples vs. Basophil: 89 samples in TestA) and is addressed via WBC Focal Loss during training (Section 4.4).

4 Methodology

4.1 Medical Enhanced Filter (MEF)

To mitigate hardware-induced image variability before learned feature extraction, we designed a five-step deterministic preprocessing pipeline.

Step 1 — Percentile-Based Color Normalization. Each RGB channel is independently stretched to the 2nd–98th percentile range:

$$I_i' = \frac{I_i - P_2^{(i)}}{P_{98}^{(i)} - P_2^{(i)}} \cdot 255, \quad i \in \{R, G, B\} \tag{1}$$

This eliminates systematic brightness differences introduced by varying microscope exposure calibrations.

Step 2 — Dual-Scale CLAHE in LAB Space. CLAHE is applied exclusively to the L (Luminance) channel after RGB-to-LAB conversion. A fine scale (4×4 tiles, clip limit 3.0) enhances granular nuclear structures; a coarse scale (8×8 tiles, clip limit 2.0) enhances broader cytoplasmic regions. Both scales are fused via a Canny edge-weighted blend:

$$L_{\text{fused}} = L_{\text{fine}}(0.5 + 0.5w_e) + L_{\text{coarse}}(0.5 - 0.5w_e) \tag{2}$$

where $w_e \in [0, 1]$ is the Gaussian-smoothed Canny edge map. Operating in LAB space prevents hue distortion of diagnostically relevant chromatin and granule coloration.

Step 3 — Edge-Preserving Bilateral Filtering. A bilateral filter (d=9, σ_{color}=65, σ_{space}=65) suppresses microscopy shot noise within cytoplasm and background while strictly preserving membrane boundaries.

Step 4 — Morphological Nucleus Highlighting. Combined inner ($k_{3\times3}$) and outer ($k_{7\times7}$) elliptical morphological gradients are additively blended to enhance nuclear boundaries:

$$G = 0.6\,\Delta_3(I_g) + 0.4\,\Delta_7(I_g) \tag{3}$$

where $\Delta_s(I_g) = \text{dilate}(I_g, k_s) - \text{erode}(I_g, k_s)$.

Step 5 — Selective Weighted Sharpening. A combined Canny and Laplacian-of-Gaussian (LoG) edge map generates an adaptive weight mask w_e. Sharpening is applied proportionally to edge confidence, preserving smooth intracellular regions:

$$I_{\text{out}} = I(1 - 0.35\,w_e) + I_{\text{sharp}}(0.35\,w_e) \tag{4}$$

4.2 Model Architecture

Backbone. DenseNet121 [1] was selected for its dense connectivity pattern, which enables parameter-efficient feature reuse—a critical property when fine-tuning on ≈12 000 images. ImageNet pre-trained weights provide the initialization.

WBCAttention. A CBAM-style [2] sequential attention module is inserted after the backbone's final feature map (7×7×1024). Channel attention summarizes each feature map via global average and max pooling, routes both through a shared two-layer MLP (reduction ratio 16, ReLU activation), and applies a sigmoid gate. Spatial attention concatenates channel-averaged and channel-max feature maps and applies a 7×7 Conv2D with sigmoid activation. The module adds 132 259 parameters (1.69% of total).

Classification Head. The attended feature map passes through GlobalAveragePooling, two Dense layers (512 and 256 units), each followed by BatchNormalization, MedSwish activation, and Dropout (rate 0.5). An auxiliary binary head provides additional supervision to disambiguate the morphologically similar Neutrophil and Lymphocyte classes during training.

MedSwish. Standard ReLU zeroes negative activations, introducing "Dying ReLU" information loss that may suppress subtle morphological gradients. MedSwish extends Swish [3] with trainable scalar parameters α and β:

$$f(x) = \underbrace{x \cdot \sigma(\beta x)}_{\text{base Swish}} \cdot \underbrace{\big(1 + \alpha \tanh(x)\big)}_{\text{bounded enhancement}} \tag{5}$$

Both α (initialized 0.1) and β (initialized 1.0) are learned jointly with the network, adding only 4 trainable scalars per layer.

The complete architecture is summarized in Table 1.

Table 1: Proposed model architecture and parameter summary.

Layer	Output Shape	Parameters
Input	$(224, 224, 3)$	0
DenseNet121	$(7, 7, 1024)$	7 037 504
WBCAttention	$(7, 7, 1024)$	132 259
GlobalAvgPool	(1024)	0
Dense 1 + BN + MedSwish + Drop	(512)	526 850
Dense 2 + BN + MedSwish	(256)	132 354
Output (Softmax)	(5)	1 285
Total		**7 830 252**

4.3 Loss Function

Given the significant class imbalance, standard cross-entropy was replaced with WBC Focal Loss [14]:

$$\mathcal{L}_{\text{focal}} = -\sum_{c=1}^{C} \alpha_c (1 - p_c)^\gamma \log(p_c) \tag{6}$$

where α_c is an inverse-frequency class weight, γ=2 is the focusing parameter, and p_c is the predicted class probability. Label smoothing (ε=0.1) was applied to reduce overconfidence.

4.4 Training Strategy

Training followed a two-phase curriculum:

- **Phase 1 (15 epochs):** DenseNet121 weights frozen; only the classification head, WBCAttention, and MedSwish parameters are trained. Adam optimizer, $\mathrm{lr} = 1 \times 10^{-3}$.
- **Phase 2 (15 epochs):** All weights unfrozen. Adam optimizer, $\mathrm{lr} = 1 \times 10^{-5}$.

Augmentation. Three stochastic augmentations were applied during training: (1) LAB-space stain jitter (per-channel shifts: luminance $\in [-8, +8]$, chrominance $\in [-6, +6]$; probability 0.3); (2) foreground-focused crop with spatial jitter and resize (probability 0.2, class-scaled to increase discrimination of Lymphocyte); (3) background randomization via Gaussian noise injection (σ=7) on estimated non-cellular regions (probability 0.15).

4.5 Dynamic Explainability Guardrail

The `XAIFocusMonitor` Keras callback integrates directly into the training loop. Every two epochs, Grad-CAM [7] heatmaps are generated for a random validation sample (n=24). The intersection of the high-activation heatmap region with the estimated cellular foreground mask is computed as a *focus ratio*. If the mean focus ratio falls below a threshold of 0.55 for three consecutive evaluation cycles, training is halted automatically. The foreground mask is estimated via a multi-cue HSV/LAB segmentation combining saturation thresholding, value thresholding, and chrominance-based nucleus hinting, followed by morphological cleanup.

4.6 Inference-Time Adaptation Pipeline

Three retraining-free techniques were composed and evaluated via incremental ablation:

1. **Binary Routing.** TestB contains only Lymphocyte and Neutrophil. A binary routing stage renormalizes the five-class softmax output to a two-class posterior before the final decision.
2. **Reinhard Stain Normalization [4].** LAB-space mean and standard deviation of each TestB image are transferred to a fixed reference derived from well-stained training images.
3. **Lightweight TTA [5].** Predictions from the original image and three augmented variants (horizontal flip, brightness perturbation, $\pm 5^\circ$ rotation) are averaged.

4.7 LLM-Based Clinical Insight Agent

At inference time, a multi-modal LLM agent receives the Grad-CAM overlay image alongside class-specific morphological context. The agent evaluates whether model attention aligns with known hematological criteria (nuclear lobation, cytoplasm granularity, chromatin density) and generates a natural language clinical insight. The architecture uses GPT-4o as primary provider with Gemini 2.5 Flash as fallback, with active provider identity exposed transparently in the web interface. When the agent reports attention focused on background structures, the inference pipeline automatically applies Reinhard normalization and re-scores the image—constituting a closed "AI supervising AI" remediation loop.

5 Experiments and Results

5.1 Backbone Selection

Five transfer learning architectures were evaluated under identical conditions—MEF preprocessing, two-phase training, same classification head—on validation accuracy and OOD generalization.

The proposed model improves Macro F1 from 0.9803 (plain DenseNet121) to 0.9853 with only 0.83 M additional parameters. It achieves the lowest inference latency (14.2 ms) despite the added components. MobileNetV2 and EfficientNetB0 underperform on morphologically fine-grained WBC subtypes, confirming that parameter efficiency does not guarantee representational capacity in medical imaging.

5.2 Class-Level Performance on TestA

Basophil achieves perfect precision and recall despite representing only 2.05% of the TestA set, demonstrating that WBC Focal Loss effectively manages extreme class imbalance. Eosinophil yields the lowest F1 (0.9517), attributable to morphological overlap with Neutrophil in cytoplasmic granulation patterns—a known challenge in hematological classification [19]. Neutrophil achieves 99.94% precision on TestB, confirming robustness of the dominant class under hardware shift.

5.3 Inference-Time Adaptation Ablation

The unadapted baseline collapses to 56.96% on TestB—marginally above random for a two-class problem—confirming severe domain shift. Binary routing contributes the largest single-stage gain (+16.94 pp) by eliminating probability mass from absent classes. Reinhard normalization provides the second-largest gain (+12.56 pp), directly confirming that staining-protocol color drift is the dominant

Table 2: Backbone comparison: validation accuracy, Macro F1, parameter count, and inference latency.

Model	Params (M)	Val. Acc. (%)	Macro F1	Inf. (ms)
VGG16 [15]	15.11	98.56	0.9724	18.1
ResNet50V2 [16]	24.75	98.17	0.9704	103.9
MobileNetV2 [17]	3.05	97.90	0.9577	96.0
EfficientNetB0 [18]	4.84	97.05	0.9418	185.4
DenseNet121 (plain) [1]	7.70	98.89	0.9803	232.2
Proposed	**7.83**	**98.53**	**0.9853**	**14.2**

Table 3: Per-class precision, recall, and F1 on the in-distribution TestA set (n=4 339).

Cell Type	Precision	Recall	F1	Count
Neutrophil	0.9962	0.9868	0.9915	2 660
Lymphocyte	0.9865	0.9884	0.9874	1 034
Basophil	**1.0000**	**1.0000**	**1.0000**	89
Monocyte	0.9372	0.9573	0.9471	234
Eosinophil	0.9265	0.9783	0.9517	322
Weighted avg	**0.9856**	**0.9853**	**0.9854**	**4 339**

Table 4: Incremental ablation of inference-time adaptation on TestA (in-distribution), TestB (OOD), and combined accuracy.

Configuration	TestA (%)	TestB (%)	Comb. (%)
Baseline (no adaptation)	97.46	56.96	84.17
+ Binary routing	97.46	73.90	89.73
+ Reinhard normalization	97.99	86.46	94.21
+ Lightweight TTA (final)	**98.53**	**89.05**	**95.42**

source of domain shift in this dataset. TTA provides a further modest +2.59 pp. The adaptation pipeline does not degrade in-distribution performance: TestA increases from 97.46% to 98.53%, confirming complementarity with the learned representations.

5.4 Cross-Backbone OOD Generalization

Table 5: All evaluated backbones on TestA (in-distribution), TestB (OOD), and combined accuracy. The *Proposed* row includes the full inference-time adaptation pipeline.

Model	TestA (%)	TestB (%)	Comb. (%)
VGG16 [15]	98.39	89.71	95.54
ResNet50V2 [16]	97.23	74.47	89.76
MobileNetV2 [17]	95.92	50.07	80.88
EfficientNetB0 [18]	96.27	61.54	84.87
DenseNet121 (plain) [1]	98.57	73.24	90.26
Proposed (final)	**98.53**	89.05	95.42

VGG16 achieves the highest raw TestB performance (89.71%), but does so with 138 M parameters—17.6× more than the proposed model. The proposed system reaches 89.05% TestB accuracy with 7.83 M parameters, demonstrating that inference-time adaptation can substitute for architectural scale in addressing domain shift. MobileNetV2 collapses on TestB (50.07%), confirming that parameter efficiency alone does not confer OOD robustness. DenseNet121 plain achieves 73.24% TestB despite its highest in-distribution validation accuracy, illustrating that training-set metrics are an unreliable proxy for clinical reliability.

5.5 MEF Preprocessing Ablation

Table 6: Ablation of preprocessing variants on the same trained model across TestA, TestB, and combined accuracy.

Preprocessing Variant	TestA (%)	TestB (%)	Comb. (%)
v1 — MEF original	98.41	85.65	94.22
v2 — Adaptive CLAHE (8×8)	97.99	**87.92**	**94.69**
v3 — v2 + top-hat / bottom-hat	95.18	77.58	89.41
v4 — v3 + Macenko (uncalibrated) [13]	57.78	42.28	52.69

v2 improves TestB by +2.27 pp over v1 at the cost of −0.42 pp on TestA, revealing a fundamental OOD–IND trade-off. The dual-scale approach in MEF (Equation 2) addresses this by fusing fine and coarse CLAHE with edge-weighted blending. v3 is unexpectedly detrimental: top-hat and bottom-hat morphological enhancement suppresses cytoplasmic texture information critical for Monocyte classification, degrading TestB by 10.34 pp. v4 represents the most severe failure: Macenko normalization applied without dataset-specific reference calibration degrades both TestA (57.78%) and TestB (42.28%) to near-random performance, confirming that reference stain vectors must be derived from target-domain statistics to be effective [13].

6 Discussion

6.1 OOD Generalization as the Primary Clinical Metric

The central finding is that in-distribution validation accuracy is an insufficient and potentially misleading indicator of clinical utility. Several evaluated backbones exceed 98% TestA accuracy yet collapse below 75% on TestB. We propose that paired in-distribution and OOD evaluation on hardware-diverse test sets should become standard reporting practice in WBC classification research.

6.2 Inference-Time Adaptation vs. Architectural Scale

The comparison between VGG16 (89.71% TestB, 138 M parameters) and the proposed system (89.05% TestB, 7.83 M parameters) demonstrates that retraining-free inference-time adaptation can approximate the OOD robustness of architecturally heavyweight models at a fraction of the computational cost.

6.3 Shortcut Learning as an Invisible Risk

The LLM agent's detection of background-focused attention despite high validation accuracy illustrates that shortcut learning is not detectable from metrics alone. The `XAIFocusMonitor` callback converts explainability from a diagnostic instrument applied after training into a preventive mechanism during training—a practically deployable approach that does not require expensive pixel-level annotations.

6.4 Limitations

Several limitations should be noted. First, the inference-time adaptation is evaluated on Raabin-WBC's specific domain shift; generalization to fluorescence microscopy or different staining chemistries has not been validated. Second, TestB contains only Lymphocyte and Neutrophil; the +32.09 pp improvement reflects performance specifically on these subtypes. Third, Macenko normalization was evaluated without dataset-specific calibration; a calibrated implementation may perform considerably better. Fourth, the LLM agent introduces API latency and external provider dependency, which may be unacceptable in air-gapped clinical environments.

7 Conclusion

We presented an end-to-end WBC classification system that treats OOD generalization as a first-class design objective. Through a combination of deterministic MEF preprocessing, a parameter-efficient custom attention and activation architecture, shortcut-resistant training with online explainability monitoring, and a three-stage retraining-free inference-time adaptation pipeline, the system achieves 98.53% in-distribution accuracy and 89.05% OOD accuracy on hardware-diverse test data. The +32.09 percentage point OOD improvement over the unadapted baseline, obtained without any model retraining, demonstrates that domain shift in hematological microscopy is substantially addressable at inference time. Ablation studies provide concrete guidance: dual-scale CLAHE outperforms single-scale variants on OOD data; morphological enhancement targeting nuclei can harm cytoplasm-dependent subtypes; and uncalibrated stain normalization is more harmful than no normalization.

Future work includes whole-slide image integration, Bayesian uncertainty calibration for OOD detection, prospective clinical validation across multiple laboratory sites, and extension to additional staining protocols via domain-specific Macenko reference calibration.

Acknowledgments

The author thanks the Department of Information Systems Engineering at Sakarya University for the academic framework within which this work was conducted. The Raabin-WBC dataset was generously provided by Tehran University under open access [11]. Computing was performed on local hardware (NVIDIA GeForce RTX 4070 Ti SUPER, AMD Ryzen 7 7800X3D, 32 GB DDR5).

Code Availability

Source code, trained weights, and evaluation scripts are available at:
`https://github.com/frissonitte/wbc-analyzer-final`

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